

Introduction to Computer Fundamentals

Short Course Material for Academic Library Use

Course Type	Easy CSE Basic Course
Level	Beginner
Prepared For	Library Management System - Academic Resource
Format	Short Notes + Key Terms + MCQ Practice

Course Learning Outcomes

- Understand basic computer parts and their functions.
- Know the difference between hardware and software.
- Learn number systems used in computers.
- Understand operating systems, files, folders, and internet basics.
- Prepare for simple quizzes and viva questions.

1. What is a Computer?

A computer is an electronic device that takes input, processes data, stores information, and produces output. It follows instructions given by programs.

Basic Computer Cycle

- Input: data given to the computer, such as keyboard typing.
- Process: CPU works on the data.
- Storage: data is saved in memory or storage devices.
- Output: result shown on monitor, printer, speaker, etc.

Examples of computers include desktop computers, laptops, smartphones, tablets, ATMs, and embedded systems inside appliances.

2. Hardware and Software

Hardware means the physical parts of a computer. Software means the programs or instructions that tell the hardware what to do.

Hardware	Software
Monitor, keyboard, mouse, CPU, RAM, hard disk	Windows, Linux, MS Word, browser, games
Can be touched physically	Cannot be touched physically
Works according to software instructions	Controls and uses hardware resources

3. Main Parts of a Computer

- CPU: The brain of the computer. It performs calculations and controls operations.
- RAM: Temporary memory used while programs are running. It is volatile.
- Storage: Permanent memory such as HDD, SSD, pen drive, or memory card.
- Motherboard: Main circuit board that connects all components.
- Input devices: Keyboard, mouse, scanner, microphone, webcam.
- Output devices: Monitor, printer, speaker, projector.

4. Memory and Storage

Memory and storage are both used to keep data, but they are not the same. RAM is fast but temporary. Storage devices are slower than RAM but can save data permanently.

RAM vs ROM

- RAM is read-write memory and loses data when power is off.
- ROM stores permanent instructions, usually for starting the computer.
- RAM is used for running programs; ROM is used for boot instructions.

Common Storage Devices

- HDD: cheaper and larger storage, but slower.
- SSD: faster and more reliable than HDD.
- Pen Drive: portable storage using USB.
- Memory Card: used in phones, cameras, and small devices.

5. Operating System

An operating system is system software that manages hardware and software resources. It works as a bridge between the user and the computer hardware.

- Examples: Windows, Linux, macOS, Android, iOS.
- Main functions: file management, memory management, process management, device management, and user interface.
- Without an operating system, normal users cannot easily run applications or control hardware.

6. Files and Folders

A file is a collection of data saved with a name. A folder is a container used to organize files. File extensions show the type of a file.

Extension	Meaning	Example
.docx	Word document	assignment.docx
.pdf	Portable document	notes.pdf
.jpg / .png	Image file	cover.jpg
.mp3	Audio file	song.mp3
.mp4	Video file	lecture.mp4

7. Number Systems

Computers use binary numbers internally because electronic circuits can easily represent two states: ON and OFF. These states are represented by 1 and 0.

System	Base	Digits Used	Example
Binary	2	0, 1	1010
Decimal	10	0 to 9	25
Octal	8	0 to 7	17
Hexadecimal	16	0 to 9 and A to F	2F

Easy Conversion Example

- Binary 1010 = $(1 \times 8) + (0 \times 4) + (1 \times 2) + (0 \times 1) = 10$ in decimal.
- Decimal 13 = 1101 in binary.

8. Internet Basics

The internet is a worldwide network of connected computers. It allows users to communicate, browse websites, send email, share files, and access online services.

- Website: a collection of web pages, such as an online library site.
- Browser: software used to visit websites, such as Chrome, Edge, or Firefox.
- URL: address of a website or web page.
- IP address: numeric address used to identify a device on a network.
- Search engine: helps find information online, such as Google or Bing.

9. Cyber Safety Basics

- Use strong passwords with letters, numbers, and symbols.
- Do not share passwords with others.
- Avoid opening unknown links and suspicious email attachments.
- Use antivirus software and keep software updated.
- Always log out from public computers after use.

10. Short Exam Notes

Important Definitions

- Computer: An electronic device that processes data according to instructions.
- Data: Raw facts and figures.
- Information: Processed and meaningful data.
- Program: A set of instructions that performs a task.
- Operating System: System software that manages computer resources.
- Input Device: A device used to enter data into a computer.
- Output Device: A device used to show results from a computer.

Common Short Questions

Q1. What is the full form of CPU?	Central Processing Unit.
Q2. What is the difference between RAM and ROM?	RAM is temporary read-write memory; ROM stores permanent instructions.
Q3. Give two examples of input devices.	Keyboard and mouse.
Q4. Give two examples of output devices.	Monitor and printer.
Q5. Why do computers use binary?	Because electronic circuits can easily represent two states: ON and OFF.

MCQ Practice

- 1. Which one is an input device? a) Monitor b) Keyboard c) Printer d) Speaker Answer: b
- 2. Which memory is volatile? a) ROM b) CD c) RAM d) SSD Answer: c
- 3. Which software manages hardware resources? a) Browser b) Operating System c) Game d) Calculator Answer: b
- 4. Binary number system has base: a) 2 b) 8 c) 10 d) 16 Answer: a
- 5. Which one is a storage device? a) Keyboard b) Mouse c) SSD d) Monitor Answer: c

End of Short Course Material

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